

16-Mei-2025								
TOPIC (CLUSTERING / REGRESSION / CLASSIFICATION / ASSOCIATION RULES)	JUDUL	DATASET (url)	CRSPDM/KDD	Preprocessing	Modeling	Evaluation	Deployment	
CLUSTERING	Prediksi Kesehatan Mental Remaja Menggunakan Perbandingan Algoritma Decision Tree, Random Forest, dan XGBoost	https://www.kaggle.com/datasets/gg-pure/hermager-mengenal-hgaly/data	CRSPDM	Missing Value Handling, Remove Duplicates, Text Cleaning, Attribute Selection, Label Encoding, TF-IDF, Data Splitting 80/20	Decision Tree, XGBoost Classifier, Random Forest, SMOTE	Silhouette Score untuk clustering, Accuracy, Precision, Recall, F1-Score, Confusion Matrix	Streamlit	
CLASSIFICATION	Analisis dan Klasifikasi Performe Akademisi Siswa Menggunakan K-Means Clustering, Logistic Regression, dan Naive Bayes Classifier	https://www.kaggle.com/datasets/yosi-clerici/students-performance-in-schools	CRISP-DM	Data Cleaning, Encoding, Data Transformation	K-Means Clustering, Logistic Regression dan Naive Bayes Classifier	Accuracy, Weighted Precision, Weighted Recall, Weighted F1-Score, Confusion Matrix	Streamlit	
CLASSIFICATION	Prediksi Fraud Detection dengan Meta Model + Model Stacking (XGBoost, Catboost, LightGBM)	EBC-CB-Fraud-Detection-1-Kaggle	CRISP-DM	Data Cleaning, Data Transformation, Feature Engineering, Feature Importance	Meta Model (Logistic Regression) + Model Stacking (XGBoost, Catboost, LightGBM)	AUC-ROC, Macro F1 Score	Streamlit	
CLUSTERING	Segmentasi Pelanggan Mail Menggunakan K-Means Clustering	https://www.kaggle.com/datasets/fnuj-siddhartha/mail-segmentation-using-k-means-clustering-dataset	CRISP-DM	Normalisasi data, pengecekan outlier, Handling Missing Value	K-Means, DBSCAN, Agglomerative Clustering	Silhouette Score, Calinski-Harabasz Index, Davies-Bouldin Index, Elbow Method	Streamlit	
Classification	DETEKSI FRAUD TRANSAKSI KARTU KREDIT MENGGUNAKAN KOMPARASI ALGORITMA RANDOM FOREST DAN NAIVE BAYES	https://www.kaggle.com/datasets/mouliknagar/randomforestfraud	CRISP-DM	Data Imputation (Median & Mode), Feature Encoding (Label Encoding & One-Hot Encoding), Data Splitting (80/20), Data Balancing (SMOTE)	Random Forest Classifier (Optimasi GridSearchCV) vs Naive Bayes (GaussianNB)	Confusion Matrix, Recall (Sensitivity) & F1-Score Kelas Minoritas, 5-Fold Cross Validation	Streamlit	
Classification	Klasifikasi Sentimen Berbasis Aspek (ABSA) pada Ulasan Wisata Alam Kabupaten Bandung Barat	https://www.kaggle.com/datasets/dezid768/natural-tourist-atractions-online-resource-download	CRISP-DM	Missing Value Handling, Remove Duplicates, Text Cleaning, Attribute Selection, Label Encoding, TF-IDF, Data Splitting 80/20	Naive Bayes, SVM, Random Forest, Logistics Regression	Accuracy, Precision, Recall, F1-Score (per aspek)	Streamlit / web app sederhana	
CLASSIFICATION	Deteksi Diri dan Klasifikasi Risiko Penyakit Jantung Menggunakan Perbandingan Algoritma Logistic Regression, Decision Tree, Random Forest, dan XGBoost	https://www.kaggle.com/datasets/jah-moushiki/heart-disease-dataset	CRISP-DM	Data Cleaning (Deduplikasi), Feature Encoding (One-Hot Encoding), Feature Scaling (StandardScaler), dan Data Splitting (80/20 dengan Stratify)	Logistic Regression, Decision Tree, Random Forest, dan XGBoost (dengan optimasi GridSearchCV)	Confusion Matrix, Recall (Sensitivity), ROC AUC, F1-Score, Accuracy, dan 5-Fold Cross Validation	Streamlit	
CLASSIFICATION	Prediksi Pembatalan Booking Hotel Menggunakan Decision Tree	https://www.kaggle.com/datasets/hy-will/hotel-reservations-classification-dataset	CRISP-DM	Data Cleaning, Encoding, Train-Test Split	Decision Tree	Confusion Matrix, Accuracy, F1-Score	Streamlit	

1 tim presentasi 15 menit + QnA 15 menit = 30 menit
4 jam
tgl 17 juni (online) pkl 11 : presentasi tubes